



HTTP Log Analysis using Splunk Project

By Sneha Deshmukh

Hands-on cybersecurity lab focused on analyzing HTTP traffic using **Splunk** and **Zeek-style JSON logs** to detect errors, anomalies, and suspicious web activity.



Objective

In this lab, you will:

- Ingest and analyze HTTP logs using **Splunk**
 - Detect **client-side (4xx)** and **server-side (5xx)** HTTP errors
 - Identify **suspicious User-Agents and URIs**
 - Detect **large file transfers** that may indicate data exfiltration
 - Gain practical experience with **SPL (Search Processing Language)**
-



Tech Stack & Tools

- **SIEM Tool:** Splunk Enterprise
 - **Log Source:** Zeek-style HTTP logs (JSON format)
 - **Index:** http_lab (custom index)
 - **Sourcetype:** json or zeek:http
-



Lab Setup

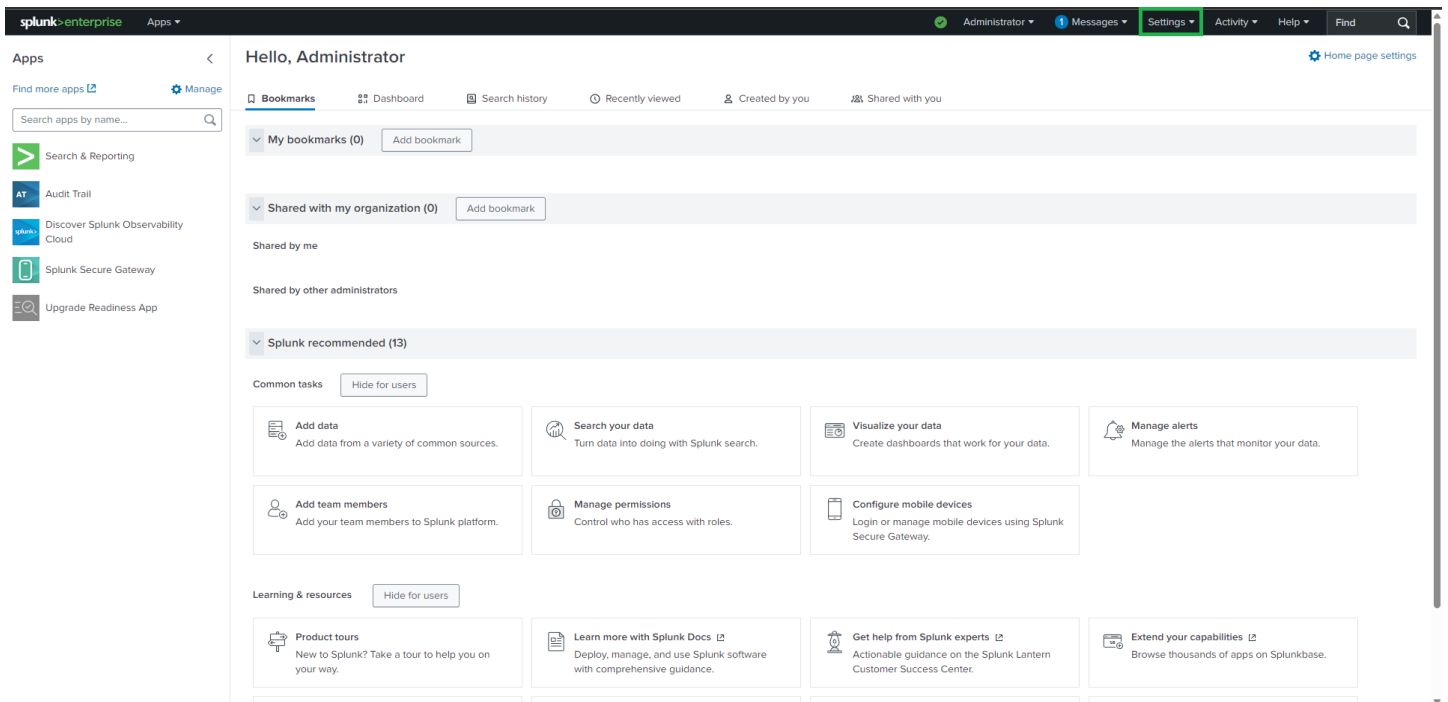


Prerequisites

- Splunk installed and accessible via Splunk Web
- HTTP logs in JSON format (http_logs.json)

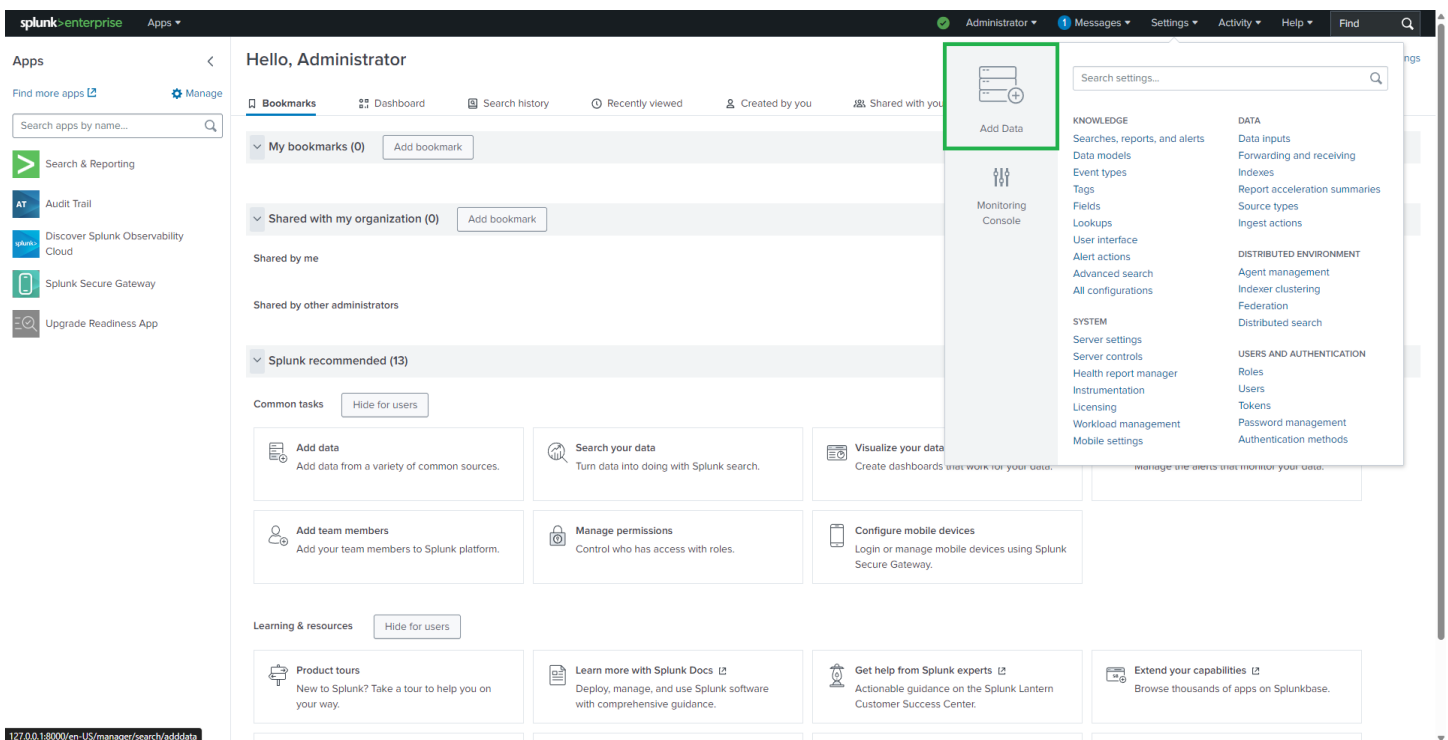
Data Ingestion Steps

1. Open Splunk Web



The screenshot shows the Splunk Web interface for an administrator. The top navigation bar includes links for Administrator, Messages, Settings (highlighted), Activity, Help, and Find. The main content area displays a 'Hello, Administrator' greeting and a list of bookmarks. Below the bookmarks, there are sections for 'My bookmarks (0)', 'Shared with my organization (0)', and 'Splunk recommended (13)'. The 'Splunk recommended' section contains a grid of common tasks and learning resources. The 'Common tasks' section includes: Add data, Search your data, Visualize your data, Manage alerts, Add team members, Manage permissions, and Configure mobile devices. The 'Learning & resources' section includes: Product tours, Learn more with Splunk Docs, Get help from Splunk experts, and Extend your capabilities.

2. Navigate to Settings → Add Data



The screenshot shows the Splunk Web interface with the 'Settings' menu open. The 'Add Data' option is highlighted with a green box. The 'Settings' menu is organized into several categories: KNOWLEDGE (Searches, reports, and alerts; Data models; Event types; Tags; Fields; Lookups; User interface; Alert actions; Advanced search; All configurations), DATA (Data inputs; Forwarding and receiving; Indexes; Report acceleration summaries; Source types; Ingest actions), DISTRIBUTED ENVIRONMENT (Agent management; Indexer clustering; Federation; Distributed search), SYSTEM (Server settings; Server controls; Health report manager; Instrumentation; Licensing; Workload management; Mobile settings), and USERS AND AUTHENTICATION (Roles; Users; Tokens; Password management; Authentication methods). The 'Add Data' option is located under the 'KNOWLEDGE' category.

3. Select Upload

splunk>enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

What data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources

Cloud computing

Get your cloud computing data in to the Splunk platform.

10 data sources

Networking

Get your networking data in to the Splunk platform.

2 data sources

Operating System

Get your operating system data in to the Splunk platform.

1 data source

Security

Get your security data in to the Splunk platform.

3 data sources

4 data sources in total

Or get data in with the following methods

Upload

files from my computer

Local log files
Local structured files (e.g. CSV)
[Tutorial for adding data](#)

Monitor

files and ports on this Splunk platform instance

Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

Forward

data from a Splunk forwarder

Files - TCP/UDP - Scripts

127.0.0.1:8000/en-US/manager/search/adddatamethods/selectforwards

splunk>enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< Back

Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: No file selected

Select File

Drop your data file here

The maximum file upload size is 500 Mb

FAQ

> What kinds of files can the Splunk platform index?

> What is a source?

> How do I get remote data onto my Splunk platform instance?

4. Upload http_logs.json

The screenshot shows the Splunk Enterprise interface during the 'Add Data' process. An 'Open' file dialog is open, showing the 'http_logs.json' file selected in the 'Downloads' folder. The 'Open' button is highlighted. The background shows the 'Add Data' wizard with the 'Select Source' step active, displaying the selected file 'http_logs.json' and a 'Drop your data file here' area. The 'Next >' button is highlighted.

Open File Dialog:

- File name: http_logs
- All Files
- Open
- Cancel

Add Data Wizard:

- Progress: Select Source (active), Set Source Type, Input Settings, Review, Done
- Buttons: < Back, Next >
- Selected File: http_logs.json
- Select File
- Drop your data file here
- The maximum file upload size is 500 Mb
- File Successfully Uploaded
- FAQ
 - > What kinds of files can the Splunk platform index?
 - > What is a source?
 - > How do I get remote data onto my Splunk platform instance?

5. Configure:

- **Source type:** json or create zeek:http
- **Index:** http_lab (recommended)

splunk>enterpriseApps

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Next >

Administrator

Messages

Settings

Activity

Help

Find

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: http_logs.json

View Event Summary

Source type: _json

Save As

> Timestamp

> Advanced

	time	event_type	id.orig_h	id.resp_h	method	resp_body_len	status_code	ts	uid	url	user_agent
1	4/25/25 4:16:18.860 PM	Large Transfer	10.0.0.49	10.0.16	GET	1958305	200	2025-04-25T10:46:18.860765Z	HT1031308	/index.html	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
2	4/25/25 4:16:18.860 PM	Unexpected Method	10.0.0.46	10.0.11	OPTIONS	2464	200	2025-04-25T10:46:18.860837Z	HT8956786	/index.html	Mozilla/5.0 (Windows NT 10.0; Win64; x64)
3	4/25/25 4:16:18.860 PM	Standard	10.0.0.24	10.0.13	POST	6163	200	2025-04-25T10:46:18.860855Z	HT9025844	/index.html	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
4	4/25/25 4:16:18.860 PM	Standard	10.0.0.30	10.0.15	GET	11580	200	2025-04-25T10:46:18.860869Z	HT8250492	/index.html	Mozilla/5.0 (X11; Linux x86_64)
5	4/25/25 4:16:18.860 PM	Large Transfer	10.0.0.13	10.0.15	POST	1223825	200	2025-04-25T10:46:18.860888Z	HT8686201	/index.html	Mozilla/5.0 (Windows NT 10.0; Win64; x64)
6	4/25/25 4:16:18.860 PM	Large Transfer	10.0.0.46	10.0.17	POST	1550707	200	2025-04-25T10:46:18.860905Z	HT7409193	/index.html	Mozilla/5.0 (X11; Linux x86_64)
7	4/25/25 4:16:18.860 PM	Standard	10.0.0.43	10.0.11	POST	7756	200	2025-04-25T10:46:18.860915Z	HT6831662	/index.html	Mozilla/5.0 (X11; Linux x86_64)
8	4/25/25 4:16:18.860 PM	Large Transfer	10.0.0.23	10.0.12	GET	1191977	200	2025-04-25T10:46:18.860930Z	HT7221204	/index.html	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
9	4/25/25 4:16:18.860 PM	Client Error	10.0.0.36	10.0.15	GET	12998	404	2025-04-25T10:46:18.860942Z	HT1157835	/index.html	Mozilla/5.0 (X11; Linux x86_64)
10	4/25/25 4:16:18.860 PM	Server Error	10.0.0.10	10.0.12	POST	1380	503	2025-04-25T10:46:18.860951Z	HT3509295	/index.html	Mozilla/5.0 (X11; Linux x86_64)
11	4/25/25 4:16:18.860 PM	Standard	10.0.0.44	10.0.11	POST	14481	200	2025-04-25T10:46:18.860960Z	HT5761373	/index.html	Mozilla/5.0 (Windows NT 10.0; Win64; x64)
12	4/25/25 4:16:18.860 PM	Standard	10.0.0.48	10.0.14	POST	4427	200	2025-04-25T10:46:18.860968Z	HT5435674	/index.html	Mozilla/5.0 (Windows NT 10.0; Win64; x64)

splunk>enterpriseApps

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Review >

Administrator

Messages

Settings

Activity

Help

Find

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Constant value

Regular expression on path

Segment in path

Host field valueLAPTOP-EHJ3QFJI

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

DefaultCreate a new index

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Input Source

Host

Index

FAQ

New Index

General Settings

Index Name

Index Data Type

Home Path

Cold Path

Thawed Path

Data Integrity Check

Max Size of Entire Index

Max Size of Hot/Warm/Cold Bucket

Frozen Path

App

Storage Optimization

Save

Cancel

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Input Source

Host

Index

FAQ

New Index

General Settings

Index Name

Index Data Type

Home Path

Cold Path

Thawed Path

Data Integrity Check

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splunk>enterpriseApps

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Select SourceSet Source TypeInput SettingsReviewDone

< BackReview >

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Default

Create a new index

Default

dns_lab

history

http_lab

main

summary

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

127.0.0.1:8000/en-US/manager/search/adddatamethods/inputsettings

splunk>enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< BackReview >

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http_lab

Create a new index

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

6. Finish upload and verify indexing

splunk>enterprise Apps

Administrator Messages Settings Activity Help Find

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Submit >

Review

Input Type Uploaded File
File Name http_logs.json
Source Type _json
Host LAPTOP-EHJ3QFJI
Index http_lab

splunk>enterprise Apps

Administrator Messages Settings Activity Help Find

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Next >

✓ File has been uploaded successfully.
Configure your inputs by going to Settings > Data Inputs

Start Searching Search your data now or see examples and tutorials. [🔗](#)

Extract Fields Create search-time field extractions. [Learn more about fields. 🔗](#)

Add More Data Add more data inputs now or see examples and tutorials. [🔗](#)

Download Apps Apps help you do more with your data. [Learn more. 🔗](#)

Build Dashboards Visualize your searches. [Learn more. 🔗](#)

Lab Tasks & SPL Queries

◆ Task 1: Top 10 Endpoints Generating Web Traffic

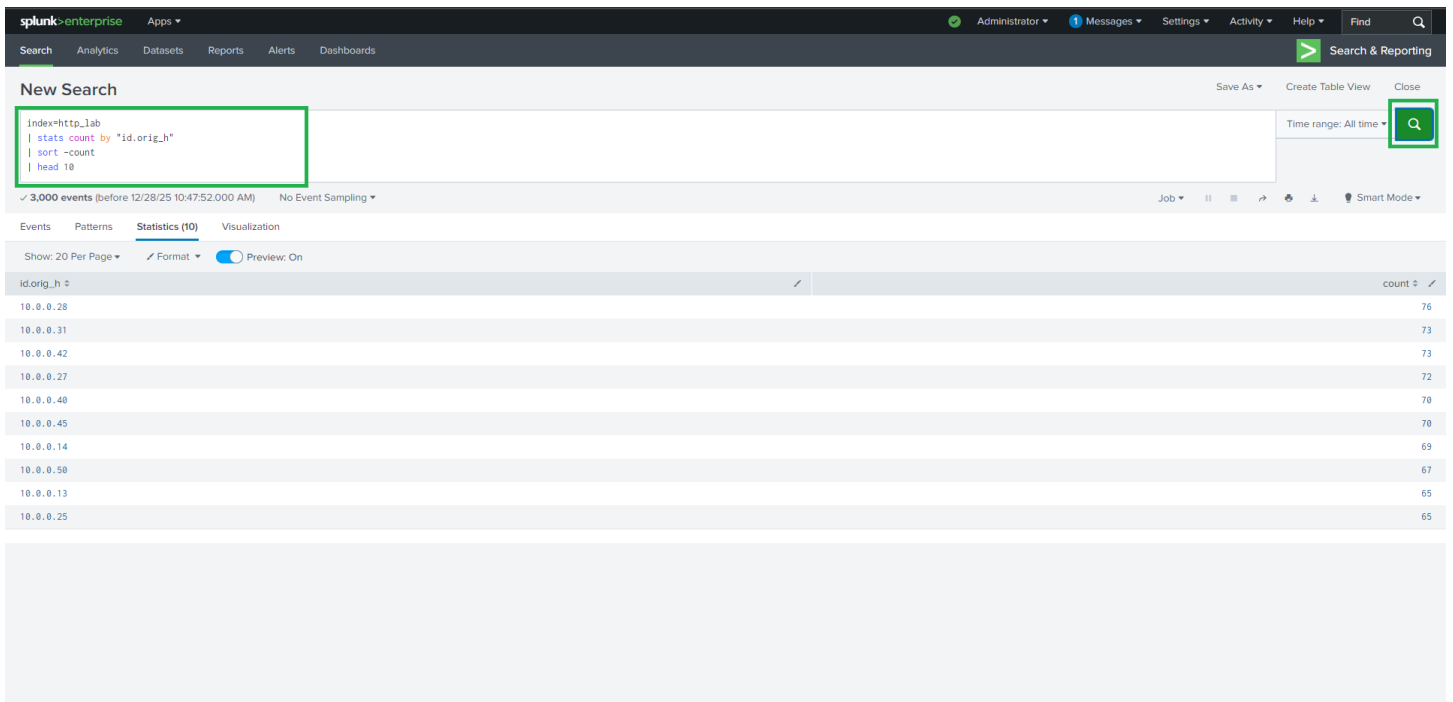
index=http_lab

| stats count by "id.orig_h"

| sort -count

| head 10

 *Identifies the most active source IPs generating HTTP requests.*



The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with 'splunk>enterprise' and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'Search & Reporting' section with a 'New Search' button. The search query is entered in a text box: `index=http_lab
| stats count by "id.orig_h"
| sort -count
| head 10`. The results are displayed in a table with 10 rows, showing the top source IPs and their corresponding counts. The table has two columns: 'id.orig_h' and 'count'. The results are sorted by count in descending order.

id.orig_h	count
10.0.0.28	76
10.0.0.31	73
10.0.0.42	73
10.0.0.27	72
10.0.0.40	70
10.0.0.45	70
10.0.0.14	69
10.0.0.58	67
10.0.0.13	65
10.0.0.25	65

◆ Task 2: Count Server Errors (HTTP 5xx)

index=http_lab status_code>=500 status_code<600

| stats count as server_errors

📌 *Helps detect backend failures or application crashes.*

The screenshot shows the Splunk Enterprise Search interface. At the top, there's a navigation bar with 'splunk>enterprise' and 'Apps'. Below it, a search bar contains the query: `index=http_lab status_code>=500 status_code<600 | stats count as server_errors`. To the right of the search bar, there's a 'Time range: All time' dropdown and a search icon. Below the search bar, it shows '285 events (before 12/28/25 10:48:35.000 AM)' and 'No Event Sampling'. At the bottom, there's a table with one row labeled 'server_errors' and a value of '285'.

splunk>enterprise Apps

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search Save As Create Table View Close

index=http_lab status_code>=500 status_code<600
| stats count as server_errors

Time range: All time

✓ 285 events (before 12/28/25 10:48:35.000 AM) No Event Sampling

Events Patterns Statistics (1) Visualization

Show: 20 Per Page Format Preview: On

server_errors
285

◆ Task 3: Detect Suspicious / Scripted User-Agents

index=http_lab

user_agent IN ("sqlmap/1.5.1", "curl/7.68.0", "python-requests/2.25.1", "botnet-checker/1.0")

| stats count by user_agent

📌 *Identifies automated tools commonly used in attacks or scanning.*

The screenshot shows the Splunk Enterprise interface. At the top, there's a navigation bar with 'splunk enterprise' logo and various menu items like 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and 'Find'. Below this is a 'Search & Reporting' section. The 'New Search' bar contains the query: `index=http_lab user_agent IN ("sqlmap/1.5.1", "curl/7.68.0", "python-requests/2.25.1", "botnet-checker/1.0") | stats count by user_agent`. The search results show 296 events. The 'Statistics (4)' tab is active, displaying a table with the following data:

user_agent	count
botnet-checker/1.0	78
curl/7.68.0	69
python-requests/2.25.1	78
sqlmap/1.5.1	79

◆ Task 4: Find Large File Transfers (> 500 KB)

index=http_lab resp_body_len>500000

| table ts "id.orig_h" "id.resp_h" uri resp_body_len

| sort -resp_body_len



Useful for spotting potential data leaks or suspicious downloads.

New Search					Save As	Create Table View	Close
Index=http_lab resp_body_len>500000 table ts "id.orig_h" "id.resp_h" uri resp_body_len sort -resp_body_len					Time range: All time		
✓ 323 events (before 12/28/25 10:49:50.000 AM) No Event Sampling					Job		Smart Mode
Events Patterns Statistics (323) Visualization							
Show: 20 Per Page Format Preview: On					< Prev 1 2 3 4 5 6 7 8 ... Next >		
ts	id.orig_h	id.resp_h	uri	resp_body_len			
2025-04-25T10:46:18.879523Z	10.0.0.23	10.0.1.4	/index.html	1977613			
2025-04-25T10:46:18.875312Z	10.0.0.49	10.0.1.7	/index.html	1976613			
2025-04-25T10:46:18.888207Z	10.0.0.50	10.0.1.3	/index.html	1974922			
2025-04-25T10:46:18.863095Z	10.0.0.30	10.0.1.5	/index.html	1968142			
2025-04-25T10:46:18.861070Z	10.0.0.48	10.0.1.12	/index.html	1968325			
2025-04-25T10:46:18.860765Z	10.0.0.49	10.0.1.6	/index.html	1958305			
2025-04-25T10:46:18.862785Z	10.0.0.57	10.0.1.10	/index.html	1951520			
2025-04-25T10:46:18.882673Z	10.0.0.36	10.0.1.7	/index.html	1945121			
2025-04-25T10:46:18.877884Z	10.0.0.15	10.0.1.10	/index.html	1944872			
2025-04-25T10:46:18.889246Z	10.0.0.35	10.0.1.2	/index.html	1941685			
2025-04-25T10:46:18.874011Z	10.0.0.43	10.0.1.11	/index.html	1929783			
2025-04-25T10:46:18.882569Z	10.0.0.28	10.0.1.5	/index.html	1924376			
2025-04-25T10:46:18.887168Z	10.0.0.32	10.0.1.4	/index.html	1916823			
2025-04-25T10:46:18.883728Z	10.0.0.51	10.0.1.9	/index.html	1909326			
2025-04-25T10:46:18.880276Z	10.0.0.40	10.0.1.8	/index.html	1902154			
2025-04-25T10:46:18.893626Z	10.0.0.36	10.0.1.5	/index.html	1901515			
2025-04-25T10:46:18.869238Z	10.0.0.57	10.0.1.8	/index.html	1893936			
2025-04-25T10:46:18.883660Z	10.0.0.35	10.0.1.1	/index.html	1889812			
2025-04-25T10:46:18.864686Z	10.0.0.21	10.0.1.12	/index.html	1883715			
2025-04-25T10:46:18.865466Z	10.0.0.36	10.0.1.7	/index.html	1879333			

◆ Task 5: Detect Suspicious URI Access Attempts

index=http_lab

uri IN ("/admin";"/shell.php";"/etc/passwd")

| stats count by uri, "id.orig_h"

🔴 *Highlights attempts to access admin panels, shells, or sensitive files.*

New Search

Save AsCreate Table ViewClose

index=http_lab uri IN ("/admin";"/shell.php";"/etc/passwd")
| stats count by uri, "id.orig_h"

98 events (before 12/28/25 10:50:35.000 AM)No Event Sampling

Time range: All time

EventsPatternsStatistics (74)Visualization

Show: 20 Per PageFormatPreview: On

uri	id.orig_h	count
/admin	10.0.0.12	1
/admin	10.0.0.14	1
/admin	10.0.0.19	1
/admin	10.0.0.20	2
/admin	10.0.0.22	1
/admin	10.0.0.25	2
/admin	10.0.0.27	1
/admin	10.0.0.28	2
/admin	10.0.0.31	1
/admin	10.0.0.34	1
/admin	10.0.0.35	2
/admin	10.0.0.36	1
/admin	10.0.0.38	2
/admin	10.0.0.39	1
/admin	10.0.0.40	1
/admin	10.0.0.42	1
/admin	10.0.0.44	1
/admin	10.0.0.46	3
/admin	10.0.0.48	1
/admin	10.0.0.49	1

🔴 Security Insights Gained

Through this lab, you can:

- Detect **malicious web reconnaissance**
- Identify **brute-force or scanning tools**
- Monitor **abnormal HTTP behavior**
- Recognize **potential data exfiltration**
- Build detection logic for **SOC use cases**

Possible Enhancements

-  Create **Splunk alerts** for:
 - Repeated 5xx errors
 - Large file transfers
 - Suspicious User-Agents
 -  Build dashboards for:
 - HTTP status trends
 - Top URIs and IPs
 -  Integrate with IDS / threat intelligence feeds
-

Learning Outcome

By completing this project, you have:

- Gained hands-on experience with **Splunk SPL**
 - Learned **real-world HTTP log analysis**
 - Practiced **threat detection techniques**
 - Strengthened **SIEM & SOC analyst skills**
-

Conclusion

This lab demonstrates how **SIEM tools like Splunk** can be effectively used to analyze web traffic, detect attacks, and improve organizational security posture using log data.

 A must-have project for **Cybersecurity, SOC Analyst, and Blue Team** portfolios.
